# Galatea: An Optimal Transport Framework for Merging Global and Local Descriptors in Visual Place Recognition on Extraterrestrial Surfaces

Raunak Sarbajna and Christoph F. Eick

***Abstract*—Visual place recognition (VPR) on the Moon has to cope with black shadows that move with the Sun, statistically self-similar crater fields and strongly non-Lambertian regolith. We present *Galatea*, a framework that treats both stages of two-stage VPR as optimal transport (OT). For global aggregation, a semi-relaxed unbalanced OT problem defines a one-parameter family of token-to-cluster assignments that contains the NetVLAD ($\lambda = 0$) and SALAD ($\lambda = 1$) assignments as end points and a learned $\lambda$ in between. We prove that its truncated Sinkhorn iterations converge linearly at rate $\lambda$, characterise the dust-bin as a logistic gate, and show that after intra-normalisation the descriptor sees the transport plan only through per-token weights, which predicts when the choice of $\lambda$ can matter. For global–local fusion, a training-free partial fused Gromov–Wasserstein matching scores both the appearance and the layout of keypoints, and a saturating fusion rule provably overrides the global ranking only where the global margin is small. On LunarSim-VPR, a physically motivated benchmark that renders the same query poses under seven Sun conditions, terrestrial models fall from about $66\%$ to $23.4\%$ Recall@1 under large Sun changes. With a frozen backbone, every value of $\lambda$ gives $\approx 59.8\%$ (zero-shot: $35.8\%$), as the theory predicts. Fine-tuning the last backbone blocks adds $11$–$19$ points; within the OT family the assignments stay within about two points and a learned $\lambda$ stays at its initial value, behaving like a fixed one, while NetVLAD-style residual aggregation pulls ahead ($76.4\%$). Fusion adds $2$–$7$ points for every stage-1 model and local matcher, because it keeps about half of the queries that local evidence rescues while breaking few that the global ranking already had right; the full system reaches $78.8\%$. Illumination-diverse training remains the largest single factor.**

***Index Terms*—Localization, visual place recognition, space robotics, optimal transport, deep learning for visual perception.**

## I. Introduction

Surface missions to the Moon need to know where they are without GNSS. Wheel and visual odometry [1] drift without bound, so a rover periodically needs an absolute fix or a loop closure. One way to get it is to recognise that the current view shows a place that is already in a reference set, either imagery from an earlier traverse or views rendered from orbital digital elevation models (DEMs) [2], [3]. This is *visual place recognition* (VPR), usually posed as image retrieval against a geo-tagged database [4], [5].

Modern VPR systems reach high recall on street-level benchmarks by aggregating deep local features into a compact global descriptor [6], [7], [8], [9], [10], optionally followed by local-feature re-ranking [11]. The lunar surface removes almost every cue these systems rely on:

- **Illumination.** There is no atmosphere to scatter light, so shadows are black and sharp. Over a lunar day the Sun's elevation and azimuth sweep through large ranges, and the same place can look entirely different at two times (Fig. 3).
- **Self-similarity.** Craters and boulders follow power-law size–frequency distributions [12], [13], so terrain is statistically repetitive at every scale and individual features are weakly discriminative [14].
- **Photometry.** Regolith is strongly non-Lambertian, with phase-dependent limb darkening and an opposition surge [15], [16], which violates the brightness-constancy assumptions built into terrestrial descriptors.
- **Data.** There is no large, pose-annotated lunar VPR dataset to train on.

Two design questions follow. *How should tokens be aggregated* when a large and variable part of every image (black sky, featureless regolith, moving shadows) carries no place information? And *how should local evidence be used* when local features respond mostly to shadow edges that move with the Sun? This paper answers both within one framework, **Galatea**, which casts each stage of two-stage VPR as an optimal-transport (OT) problem. Global aggregation becomes an assignment of tokens to clusters and a dust-bin; local verification becomes a transport between keypoint sets that respects their layout; and the two are combined by a fusion rule whose behaviour can be stated exactly. Treating the stages as OT problems makes their design choices explicit and analysable: the framework contains the assignments of NetVLAD and SALAD as special cases, its theory predicts when the choice between them matters, and our experiments test those predictions with a frozen and with a fine-tuned backbone. Our contributions are:

1) **An OT family of aggregators** (Sec. IV-B). A semi-relaxed unbalanced OT problem defines token-to-cluster assignments indexed by one scalar $\lambda \in [0, 1]$ that recovers the NetVLAD soft-assignment ($\lambda = 0$) and the balanced Sinkhorn assignment of SALAD [10] ($\lambda = 1$) *exactly*; Galatea can also learn $\lambda$. We prove linear convergence with rate $\lambda$ for the truncated iterations, characterise the dust-bin as a logistic gate on cluster affinity, and show that intra-normalised descriptors see the transport plan only through per-token weights

The authors are with the Department of Computer Science, University of Houston, Houston, TX, USA.

(Props. 1 to 4).

2) **OT-based global–local fusion** (Secs. IV-E and IV-F). A training-free partial fused Gromov–Wasserstein (FGW) matcher scores both the appearance and the relative layout of keypoints and is invariant to similarity transforms (Prop. 5). A saturating fusion rule combines it with the global similarity and provably overturns the global ranking only where its margin is small (Prop. 6).
3) **LunarSim-VPR** (Sec. V)[1], an open, procedurally generated benchmark with a physically motivated renderer (crater size–frequency statistics, lunar-Lambert photometry with opposition surge, cast shadows from local and regional relief) whose protocol separates viewpoint change from controlled changes in solar elevation and azimuth.
4) **A controlled study of what matters** (Secs. VI and VII). Under identical training, with three seeds and paired-bootstrap confidence intervals, we evaluate terrestrial models zero-shot, the whole assignment family with a frozen and a fine-tuned backbone, four local scorers with and without fusion, and the effect of training illumination. We distil the results into design guidelines for lunar VPR.

The findings are not all in favour of the most flexible model. With a frozen backbone every value of $\lambda$ performs alike, exactly as Prop. 4 predicts. With a fine-tuned backbone the family spreads only slightly, a learned $\lambda$ is no better than fixing it at its initial value, and residual NetVLAD aggregation, a different design axis, pulls ahead. Fusion, in contrast, helps every stage-1 model, and how local and global evidence are combined matters more than which local matcher is used.

## II. Related Work

*Global descriptors for VPR:* VLAD [17] and its trainable counterpart NetVLAD [6] softly assign local features to clusters and sum their residuals. GeM pooling [7] and classification-based training at scale (CosPlace [8], EigenPlaces [9]) give strong compact descriptors. MixVPR [18] mixes feature maps with MLPs. AnyLoc [19] showed that unsupervised aggregation of DINOv2 [20] features generalises across domains, including aerial and underwater imagery. SALAD [10] recast NetVLAD's assignment as balanced entropic OT with a dust-bin on top of a partially fine-tuned DINOv2 backbone. Our aggregator generalises SALAD to unbalanced OT and contains NetVLAD's assignment as a limiting case.

*Re-ranking and matching:* Two-stage pipelines re-rank global candidates with local evidence, for example multi-scale patch descriptors [11] or keypoint matching with geometric verification. Learned matchers such as SuperGlue [21] and LightGlue [22] solve (or approximate) an OT problem with dust-bins over attention-refined SuperPoint [23] features. They are trained on terrestrial imagery and assume that the descriptors carry enough appearance information. Our re-ranker is training-free. It adds a Gromov–Wasserstein structural term [24], [25], in the fused form of [26] and with partial mass [27], so that matches must preserve relative keypoint geometry.

*Optimal transport:* Entropic regularisation and the Sinkhorn algorithm [28], [29] made OT differentiable and fast. Unbalanced OT replaces hard marginal constraints with divergence penalties and keeps Sinkhorn-like scaling algorithms [30]. Gromov–Wasserstein distances compare metric spaces through their internal distances [24] and are used for structured matching [26]. We use OT in both roles: as a layer inside the descriptor and as a training-free matcher between keypoint sets.

*Planetary localisation:* Rovers rely on visual odometry [1], and landers on terrain-relative navigation against orbital maps [2]. Synthetic optical images rendered from DTMs support lunar navigation [3]. Our DESPINA [31] synthesises photorealistic horizon views for a given location and heading with DEM-conditioned diffusion. Lunar crater images have been retrieved with visual and depth features [32]. Surface-level place recognition under controlled illumination change remains largely unstudied.

## III. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{I}_r^{\mathrm{db}}, \mathbf{x}_r, \psi_r)\}_{r=1}^{N}$ be a database of reference images with known planar position $\mathbf{x}_r \in \mathbb{R}^2$ and heading $\psi_r$, and let $\mathbf{I}^q$ be a query image taken at an unknown pose $(\mathbf{x}^q, \psi^q)$, generally under a different illumination. A database image is a *true match* of the query if $\|\mathbf{x}_r - \mathbf{x}^q\|_2 < \rho$ and $|\psi_r - \psi^q|_\circ < \vartheta$, where $|\cdot|_\circ$ is the wrapped angular difference (we use $\rho = 10\,\mathrm{m}$, $\vartheta = 60^\circ$). VPR is posed as retrieval [4], [5]: an encoder $\Phi$ maps every image to a unit-norm global descriptor and the database is ranked by cosine similarity $s_g(q, r) = \langle \Phi(\mathbf{I}^q), \Phi(\mathbf{I}_r^{\mathrm{db}}) \rangle$. A two-stage system additionally re-scores the top-$K$ candidates with a local score $s_\ell(q, r)$ computed from keypoint correspondences. Performance is measured by Recall@$k$, the fraction of queries with at least one true match among the top-$k$ results:

$$\mathrm{R@}k = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}\big[\exists\, r \in \mathrm{top}_k(q) :\ r \text{ is a true match of } q\big]. \tag{1}$$

## IV. The Galatea Framework

Fig. 1 gives an overview. Galatea has three OT components with explicit design parameters. Stage 1 aggregates the patch tokens of a vision transformer into a global descriptor by solving a *semi-relaxed unbalanced* optimal-transport (UOT) problem between tokens and a set of learned clusters plus a dust-bin (Sec. IV-B); its relaxation $\lambda$ selects a member of an assignment family that includes NetVLAD and SALAD. Stage 2 re-ranks the $K$ best candidates with a *partial fused Gromov–Wasserstein* (FGW) matching of keypoints, which scores both what the keypoints look like and how they are arranged (Sec. IV-E). A saturating rule fuses the global and local scores (Sec. IV-F). The backbone may be frozen or fine-tuned jointly with the head; the theory below holds in both cases, but its consequences differ (Sec. IV-C).

[1]Code and data: https://github.com/RaunakDune/galatea-lunar-vpr

### A. Tokens, scores and the assignment problem

A DINOv2 ViT-B/14 backbone [20] produces $n$ patch tokens $\mathbf{t}_1, \dots, \mathbf{t}_n \in \mathbb{R}^d$ and a class token $\mathbf{t}_{n+1}$ ($d = 768$; $n = 300$ for a $210 \times 280$ input). Following [10], two small MLPs map each token to an assignment score vector $\mathbf{s}_i = (s_{i1}, \dots, s_{im}) \in \mathbb{R}^m$ over $m$ clusters and to a reduced feature $\mathbf{f}_i \in \mathbb{R}^\ell$:

$$\begin{aligned} \mathbf{s}_i &= \mathbf{W}_{s_2}\, \sigma(\mathbf{W}_{s_1}\mathbf{t}_i + \mathbf{b}_{s_1}) + \mathbf{b}_{s_2}, \\ \mathbf{f}_i &= \mathbf{W}_{f_2}\, \sigma(\mathbf{W}_{f_1}\mathbf{t}_i + \mathbf{b}_{f_1}) + \mathbf{b}_{f_2}. \end{aligned} \tag{2}$$

Unlike NetVLAD [6], the score layer is randomly initialised (no $k$-means centroids), so no geometric prior on the clusters is imposed. A learnable scalar $z$ scores every token against an extra *dust-bin* column, giving the augmented score matrix $\bar{\mathbf{S}} = [\mathbf{S} \,|\, z\mathbf{1}_n] \in \mathbb{R}^{n\times(m+1)}$. Each token carries unit mass, $\mathbf{a} = \mathbf{1}_n$, and the clusters and dust-bin have nominal capacities $\mathbf{b} = (\mathbf{1}_m^\top, n-m)^\top$ (this requires $n > m$). We write $H(\mathbf{P}) = -\sum_{ij} P_{ij}(\log P_{ij} - 1)$ for the entropy and $\mathrm{KL}(\mathbf{x}\|\mathbf{y}) = \sum_j x_j \log(x_j/y_j) - x_j + y_j$.

*a) Two known special cases:* NetVLAD distributes each token's mass independently over the clusters with a softmax, $P_{ij} = \mathrm{softmax}_j(s_{ij}/\varepsilon)$. This constrains only the rows of $\mathbf{P}$ (feature→cluster) and ignores how much mass each cluster receives (cluster→feature). SALAD [10] constrains both marginals by solving the balanced entropic OT problem [28]

$$\mathbf{P}^{\mathrm{bal}} = \underset{\mathbf{P}\geq 0,\ \mathbf{P}\mathbf{1}=\mathbf{a},\ \mathbf{P}^\top\mathbf{1}=\mathbf{b}}{\arg\max} \langle \bar{\mathbf{S}}, \mathbf{P}\rangle + \varepsilon H(\mathbf{P}). \tag{3}$$

At convergence, with $n = 300$ and $m = 64$, (3) puts exactly one unit of mass into every cluster and $n - m = 236$ units ($79\%$ of all mass) into the dust-bin, regardless of image content. A priori this is a questionable prior on the lunar surface, where the fraction of uninformative tokens (black sky, featureless regolith, saturated slopes) varies widely with the horizon position and the Sun elevation. In practice SALAD truncates Sinkhorn after $L = 3$ iterations, which relaxes the constraint somewhat (Sec. VI-C). The formulation below makes the relaxation explicit and controllable.

### B. A family of semi-relaxed unbalanced OT assignments

We keep the token constraint hard (every token distributes exactly its unit mass) but replace the cluster/dust-bin constraint with a KL penalty of strength $\tau > 0$ [30]:

$$\mathbf{P}^\tau = \underset{\mathbf{P}\geq 0,\ \mathbf{P}\mathbf{1}_{m+1}=\mathbf{a}}{\arg\max} \langle \bar{\mathbf{S}}, \mathbf{P}\rangle + \varepsilon H(\mathbf{P}) - \tau\,\mathrm{KL}\big(\mathbf{P}^\top\mathbf{1}_n \,\big\|\, \mathbf{b}\big). \tag{4}$$

Define $\lambda = \tau/(\tau+\varepsilon) \in (0,1)$ and the log-sum-exp $\mathrm{LSE}_j(x_j) = \log\sum_j e^{x_j}$. Proofs of Props. 1 to 5 are given in the supplementary material.

**Proposition 1** (Structure, limits)**.** *Problem* (4) *has a unique solution of the form* $P^\tau_{ij} = \exp\big((\bar{S}_{ij} + f_i + g_j)/\varepsilon\big)$*, where the potentials* $(\mathbf{f}, \mathbf{g})$ *satisfy*

$$f_i = \varepsilon \log a_i - \varepsilon\,\mathrm{LSE}_j\big((\bar{S}_{ij} + g_j)/\varepsilon\big), \tag{5}$$

$$g_j = \lambda\big[\varepsilon \log b_j - \varepsilon\,\mathrm{LSE}_i\big((\bar{S}_{ij} + f_i)/\varepsilon\big)\big]. \tag{6}$$

*(i) As* $\tau \to 0$ *(*$\lambda \to 0$*),* $\mathbf{g} \to \mathbf{0}$ *and* $P^\tau_{ij} \to a_i\,\mathrm{softmax}_j(\bar{S}_{ij}/\varepsilon)$*: the NetVLAD soft-assignment, augmented with a dust-bin. (ii) As* $\tau \to \infty$ *(*$\lambda \to 1$*),* (5)–(6) *become the balanced Sinkhorn fixed-point equations and* $\mathbf{P}^\tau \to \mathbf{P}^{\mathrm{bal}}$ *of* (3)*.*

Hence a single scalar $\lambda$ interpolates continuously between NetVLAD-style and SALAD-style assignment. $\lambda$ can be fixed, or *learned* as $\lambda = \mathrm{sigmoid}(\theta)$ jointly with the rest of the head; we call the learned variant the Galatea head. Alternating (5) and (6) gives Alg. 1, which is SALAD's Sinkhorn loop with the cluster-side update multiplied by $\lambda$; it costs $O(Lnm)$.

**Proposition 2** (Linear convergence)**.** *Let* $\Phi$ *map* $\mathbf{g}$ *to the value of* (6) *after substituting* $\mathbf{f}(\mathbf{g})$ *from* (5)*. Then* $\|\Phi(\mathbf{g}) - \Phi(\mathbf{g}')\|_\infty \leq \lambda\|\mathbf{g} - \mathbf{g}'\|_\infty$*. For* $\lambda < 1$ *the iterates of Alg. 1 therefore converge linearly,* $\|\mathbf{g}^{(L)} - \mathbf{g}^\star\|_\infty \leq \lambda^L\|\mathbf{g}^{(0)} - \mathbf{g}^\star\|_\infty$*, at a rate independent of* $\varepsilon$*,* $n$ *and* $m$*.*

For the balanced case ($\lambda = 1$) the same map is only non-expansive in $\|\cdot\|_\infty$. Its convergence is usually proved in Hilbert's projective metric, with a rate that depends on $\varepsilon$ and on the dynamic range of $\bar{\mathbf{S}}$ [29]. Prop. 2 instead gives an explicit, data-independent rate for every $\lambda < 1$. Numerically the bound is tight: on random score matrices the measured error ratio between successive iterations approaches $\lambda$. With learned values of $\lambda$ (Sec. VI), a small fixed $L$ is therefore justified.

**Proposition 3** (Dust-bin gating)**.** *For every* $\lambda \in [0,1]$*, after any token-side update* (5) *(in particular at the output of Alg. 1), the fraction of token* $i$*'s mass sent to the dust-bin is*

$$P_{i,m+1} = \mathrm{sigmoid}\Big(\tfrac{1}{\varepsilon}\big[z + g_{m+1} - \varepsilon\,\mathrm{LSE}_{j\leq m}\big((s_{ij} + g_j)/\varepsilon\big)\big]\Big), \tag{7}$$

*which, for fixed potentials* $\mathbf{g}$*, is strictly decreasing in every cluster score* $s_{ij}$ *and increasing in* $z$*.*

The dust-bin therefore acts as a learned soft gate on the token's *soft-max cluster affinity*. Tokens that resemble no cluster, such as sky, smooth regolith or noise, are suppressed before aggregation. In the relaxed problem, the potential $g_{m+1}$, and with it the total discarded mass, adapts to the image instead of being fixed at $n - m$.

The next result shows which part of the transport plan the descriptor actually sees. It explains the empirical behaviour reported in Sec. VI-F.

**Proposition 4** (Column-scale invariance)**.** *Let the descriptor be built from* $\mathbf{P}$ *by* (9) *followed by per-cluster (intra-)* $\ell_2$*-normalisation. Then* $\Phi$ *is invariant to* $\mathbf{P} \mapsto \mathbf{P}\,\mathrm{diag}(\mathbf{c})$ *for any* $\mathbf{c} > 0$*. In particular, for any plan of the form of Prop. 1,* $\Phi$ *depends on the cluster potentials* $\mathbf{g}_{1:m}$ only through *the token weights* $w_i = e^{f_i/\varepsilon} = a_i \big/ \sum_{j\leq m+1} e^{(\bar{S}_{ij}+g_j)/\varepsilon}$*:*

$$\frac{\mathbf{V}_{j,:}}{\|\mathbf{V}_{j,:}\|} = \frac{\sum_i w_i e^{s_{ij}/\varepsilon}\mathbf{f}_i}{\big\|\sum_i w_i e^{s_{ij}/\varepsilon}\mathbf{f}_i\big\|}. \tag{8}$$

Balanced and unbalanced assignments therefore differ only in how they *weight tokens*, through $w_i$. How much total mass each cluster receives is removed by intra-normalisation. Equation (8) also shows that NetVLAD with a dust-bin ($\lambda = 0$) is SALAD-style aggregation with the simplest weights, $w_i = 1/(\sum_j e^{s_{ij}/\varepsilon} + e^{z/\varepsilon})$.

**Algorithm 1** $\lambda$-scaled log-domain Sinkhorn (Galatea assignment)

**Require:** scores $\bar{\mathbf{S}} \in \mathbb{R}^{n\times(m+1)}$, marginals $\mathbf{a}, \mathbf{b}$, $\lambda \in [0,1]$, iterations $L$

$\mathbf{f} \leftarrow \mathbf{0}_n,\ \mathbf{g} \leftarrow \mathbf{0}_{m+1}$ ▷ $\varepsilon$ absorbed in $\bar{\mathbf{S}}$

**for** $\ell = 1, \dots, L$ **do**

  $g_j \leftarrow \lambda\left[\log b_j - \mathrm{LSE}_i(\bar{S}_{ij} + f_i)\right]$ ▷ clusters + dust-bin (relaxed)

  $f_i \leftarrow \log a_i - \mathrm{LSE}_j(\bar{S}_{ij} + g_j)$ ▷ tokens (hard)

**return** $\mathbf{P} = \exp(\bar{\mathbf{S}} + \mathbf{f}\mathbf{1}^\top + \mathbf{1}\mathbf{g}^\top)$ with the dust-bin column removed

### C. What the theory predicts

Props. 2 and 4 make three testable predictions. (P1) With fixed tokens, members of the family differ only through a re-weighting $w_i$ of tokens that already pass the dust-bin gate, a second-order effect, so all values of $\lambda$ should give similar retrieval accuracy. (P2) For the same reason the loss is nearly flat in $\lambda$, so a learned $\lambda$ should stay close to its initialisation. (P3) Because the iteration contracts at rate $\lambda$, a few Sinkhorn steps should suffice, and their number should not matter. None of these statements constrains what happens when the tokens themselves are trained: a fine-tuned backbone can adapt its tokens to whichever weighting the assignment imposes, so the family may separate. The theory also says nothing about the *aggregation* (residuals to centroids as in NetVLAD, or SALAD's reduced features), which is a second, independent design axis. Sec. VI-C tests both regimes.

### D. Aggregation and training

With $\mathbf{P} \in \mathbb{R}^{n\times m}$ (the dust-bin column dropped), the aggregated descriptor is the matrix $\mathbf{V} \in \mathbb{R}^{m\times\ell}$ with entries

$$V_{j,k} = \sum_{i=1}^{n} P_{i,j}\, f_{i,k}, \qquad j = 1,\dots,m, \quad k = 1,\dots,\ell. \tag{9}$$

As in [10], centroids are not subtracted. A scene descriptor $\mathbf{g}_{\mathrm{sc}} = \mathbf{W}_{g_2}\sigma(\mathbf{W}_{g_1}\mathbf{t}_{n+1} + \mathbf{b}_{g_1}) + \mathbf{b}_{g_2}$ summarises the class token. Each row of $\mathbf{V}$ is $\ell_2$-normalised (intra-normalisation), $\mathbf{g}_{\mathrm{sc}}$ is $\ell_2$-normalised, and the concatenation $[\mathbf{g}_{\mathrm{sc}}; \mathrm{vec}(\mathbf{V})]$ is $\ell_2$-normalised again to give $\Phi(\mathbf{I}) \in \mathbb{R}^{256+m\ell}$ (8448-D for $m = 64, \ell = 128$). We train with the multi-similarity loss and miner [33] on batches of $P$ places $\times$ 4 views. Images of *different* places closer than 15 m are masked out of the negative set, because adjacent places along a traverse overlap visually.

### E. Stage 2: partial fused Gromov–Wasserstein re-ranking

Repetitive crater and boulder fields make individual local descriptors ambiguous: two places may share many similar-looking craters. What distinguishes them is *where* those craters lie relative to one another [14]. Let $\{(\mathbf{u}_i, \mathbf{d}_i)\}_{i=1}^{p}$ and $\{(\mathbf{v}_k, \mathbf{e}_k)\}_{k=1}^{p'}$ be SuperPoint keypoints [23] (pixel positions and unit descriptors) of the query and a candidate. We define the appearance cost $M_{ik} = 1 - \langle \mathbf{d}_i, \mathbf{e}_k\rangle$ and scale-normalised intra-image distance matrices $C^q_{ii'} = \|\mathbf{u}_i - \mathbf{u}_{i'}\|/\mu_q$ and $C^c_{kk'} = \|\mathbf{v}_k - \mathbf{v}_{k'}\|/\mu_c$, where $\mu$ is the mean pairwise distance in each image. The partial FGW problem [26], [27] trades appearance against preservation of the keypoint geometry:

$$\min_{\mathbf{T}\in\bar{\mathcal{U}}} (1-\alpha)\langle \mathbf{M}, \mathbf{T}\rangle + \alpha \sum_{i,i',k,k'} \left(C^q_{ii'} - C^c_{kk'}\right)^2 T_{ik} T_{i'k'} - \varepsilon_\ell H(\bar{\mathbf{T}}), \tag{10}$$

where $\bar{\mathbf{T}} \in \mathbb{R}^{(p+1)\times(p'+1)}$ augments $\mathbf{T}$ with dust-bins that absorb non-co-visible keypoints [21], and $\bar{\mathcal{U}}$ fixes unit mass per keypoint, with $p'$ and $p$ units for the bins. We minimise (10) by entropic mirror descent [25]. Each step linearises the quadratic term at the current transported part $\mathbf{T}^{(r)}$ (normalised to unit mass, with marginals $\boldsymbol{\pi}, \boldsymbol{\pi}'$):

$$\mathbf{G}^{(r)} = (\mathbf{C}^q)^{\odot 2}\boldsymbol{\pi}\mathbf{1}^\top + \mathbf{1}\boldsymbol{\pi}'^\top(\mathbf{C}^c)^{\odot 2} - 2\,\mathbf{C}^q\mathbf{T}^{(r)}\mathbf{C}^{c\top}. \tag{11}$$

(this is the gradient of the quadratic term up to a factor 2, absorbed into $\alpha$). It then solves an entropic OT problem with dust-bins on the fused cost $(1-\alpha)\mathbf{M} + \alpha\mathbf{G}^{(r)}$ with the log-domain Sinkhorn algorithm. The initial plan uses $\mathbf{M}$ alone. Correspondences are mutual arg-maxima of $\mathbf{T}$ with confidence above 0.2. They are verified with a MAGSAC++ [34] fundamental-matrix fit, and the local score $s_\ell$ is the number of inliers.

**Proposition 5** (Invariances of the structural term)**.** *The objective* (10)*, and hence $s_\ell$ before geometric verification, is invariant to any similarity transformation (rotation, translation, isotropic scaling) applied to the keypoint positions of either image, and to any permutation of the keypoints.*

Unlike homography- or epipolar-based scoring, the structural term thus needs no camera model. It rewards candidates in which matched keypoints keep their *relative* layout, which is exactly the cue that separates two crater fields with similar statistics.

### F. Fusing global and local evidence

The final score of candidate $r$ for query $q$ is

$$s(q,r) = s_g(q,r) + \beta\,\frac{s_\ell(q,r)}{s_\ell(q,r) + \kappa}, \tag{12}$$

where the saturating map $\phi(s) = s/(s+\kappa) \in [0,1)$ keeps one very textured candidate from dominating. $\beta$ and $\kappa$ are selected on the validation terrain. With $\beta \to \infty$ the method reduces to pure local re-ranking, and with $\beta = 0$ to the global ranking. The fusion rule treats the global similarity margin as a confidence signal:

**Proposition 6** (Bounded override)**.** *Let $r, r'$ be two candidates of query $q$ with $s_g(q,r') - s_g(q,r) = \delta \geq 0$. Under* (12)*, $r$ is ranked above $r'$ only if $\delta < \beta\,[\phi(s_\ell(q,r)) - \phi(s_\ell(q,r'))] < \beta$. Hence the fused ranking agrees with the global ranking on every pair whose global margin is at least $\beta$, and with the local ranking on every pair whose local scores are equal or ordered like the global ones.*

*Proof.* $s(q,r) > s(q,r')$ is equivalent to $\beta[\phi(s_\ell(q,r)) - \phi(s_\ell(q,r'))] > \delta$, and $\phi \in [0,1)$. If $s_\ell(q,r) \leq s_\ell(q,r')$ the left side is non-positive, since $\phi$ is increasing. □

Pure local re-ranking has no such guard: one candidate with many shadow-edge matches overturns even a confident global decision. Sec. VI-E shows that this is exactly how local-only re-ranking loses accuracy on the Moon, and how fusion avoids it.

### G. Complexity

Alg. 1 costs $O(Lnm)$ (about $6 \cdot 10^4$ operations for $L = 3$), which is negligible next to the ViT forward pass. Each FGW step costs $O(p^2p' + pp'^2)$ for (11) plus $O(pp')$ per Sinkhorn iteration. Re-ranking $K$ candidates with $R$ outer steps therefore costs $O(KRp^3)$ for $p \approx p'$. Measured run times are reported in Sec. VI-G.

## V. The LunarSim-VPR Benchmark

No public lunar surface dataset combines dense pose ground truth with systematic illumination coverage. We therefore built a lightweight, reproducible renderer that targets the three properties that make lunar VPR hard: self-similar relief, hard shadows and non-Lambertian photometry. It does not replace real data (Sec. VII-A), but it lets us vary one factor at a time with exact ground truth. It complements DESPINA [31], which produces realistic appearance for real sites from LRO DEMs but does not expose explicit control of the Sun direction.

*Terrain:* Each scene combines a *regional* DEM (16 m/px, 16.4 km) and a *local* DEM (0.3 m/px, 614 m) blended into its centre. Both start from fractional-Brownian relief (power spectrum $\propto k^{-\beta}$, $\beta \in [2.2, 3.2]$). Craters are drawn from an equilibrium-like cumulative size–frequency distribution $N(>D) = cD^{-2}$ [13] ($D \in [60, 2500]$ m regionally and $[1.5, 120]$ m locally) and stamped with a bowl–rim–ejecta profile whose depth-to-diameter ratio (0.04–0.2) and rim height depend on a random degradation state. Fresh craters get bright albedo halos and ejecta boulders. Four to seven distant massifs (200–700 m high, 4–7.5 km away) give a horizon silhouette, as at the Apollo 17 site (Fig. 2).

*Photometry and shadows:* Radiance follows the lunar-Lambert function of McEwen [15], $I \propto A\,[2L(g)\cos i/(\cos i + \cos e) + (1 - L(g))\cos i]$, with phase-dependent $L(g)$. It is multiplied by a Hapke-style shadow-hiding opposition term $1+B_0/(1+\tan(g/2)/h)$ [16], where $i$, $e$ and $g$ are the incidence, emission and phase angles. Cast shadows are computed from both DEMs (so massifs shade the valley at low Sun), averaged over the $0.53^\circ$ solar disk to produce penumbrae. A weak ambient term models earthshine and inter-reflection, and a footprint-faded micro-relief bump map adds sub-grid regolith texture.

*Camera:* A rover-mounted pinhole camera (70° HFOV, $320 \times 240$, 1.35–1.7 m high) is rendered by column-wise height-field ray casting, accounting for lunar curvature. The renderer applies centre-weighted auto-exposure, a soft-shoulder tone curve, gamma, shot and read noise, and 8-bit quantisation. One image takes about 0.15 s on a CPU once the shadow maps for a Sun direction have been computed.

TABLE I
Query illumination conditions (Sun azimuth/elevation in degrees). The database is rendered at azimuth 135°, elevation 30°.

| | C0 | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|---|
| Az./El. | 135/30 | 135/15 | 135/6 | 135/60 | 225/30 | 315/30 | 315/8 |
| Change | view | lower | grazing | high | rot. 90° | flip | flip+graz. |

*Protocol:* Each scene contains a smooth random traverse with places every 6 m. The database is rendered at one Sun direction (azimuth 135°, elevation 30°). Queries are taken at jittered poses (up to 3 m offset, $\mathcal{N}(0, 10^\circ)$ heading noise, random camera height). The *same* query poses are rendered under seven Sun conditions (Table I), so differences between conditions are paired and caused by illumination alone. The test set has three unseen scenes ($3 \times 150$ database images and $3 \times 1050$ queries), the validation set one scene (100 places, 700 queries), and the training set five further scenes with 240 places $\times$ 4 views. Training views use Sun directions drawn from a per-scene pool of 16 random directions (elevation 4°–75°, any azimuth), which mimics an illumination-sweep dataset.

## VI. Experiments

### A. Setup

*Compared methods: Zero-shot terrestrial models* use their official weights: CosPlace-R50 [8] (512-D), EigenPlaces-R50 [9] (2048-D) and DINOv2-SALAD [10] (8448-D, trained on GSV-Cities [36]). *Lunar-trained heads* all start from the SALAD-fine-tuned DINOv2 ViT-B/14 backbone and share the training data, the loss and the schedule; they differ only in the aggregation. Besides GeM [7] (1280-D) and NetVLAD [6] (residuals to $k$-means-initialised centroids, $64 \times 128$ + scene token, 8448-D), we evaluate members of the OT family of Sec. IV-B, which share SALAD's aggregation and differ only in the assignment: $\lambda = 0$ (softmax + dust-bin), fixed $\lambda = 0.25$ and 0.75, learned $\lambda$ (the Galatea head, initialised at 0.5) and $\lambda = 1$ (SALAD), each with and without the dust-bin at the end points. All 8448-D heads have almost identical parameter counts. Each head is trained either on a *frozen* backbone or *fine-tuned* jointly with the last four ViT blocks, as in SALAD. For stage 2 we compare four local scorers on the same SuperPoint keypoints (top 384 by score) and top-$K = 10$ candidates: mutual nearest neighbours with cosine $> 0.7$ (MNN), training-free entropic OT with dust-bins (UOT, appearance only), the learned LightGlue matcher [22], and our partial FGW. Each is used with and without MAGSAC++ verification.

*Implementation:* Images are resized to $210 \times 280$ for the ViT ($n = 300$ tokens). Heads are trained with AdamW (learning rate $4 \cdot 10^{-4}$, one-cycle schedule, weight decay $10^{-4}$) for 500 steps of 24 places $\times$ 4 views, using the multi-similarity loss ($\alpha = 1$, $\beta = 50$, miner margin 0.1). Sinkhorn uses $L = 3$ iterations, $m = 64$ clusters and $\ell = 128$. All heads in Tables II and III are run with three seeds (mean $\pm$ standard deviation); the ablations of Table VI use one seed. Confidence intervals are 95% paired bootstraps over

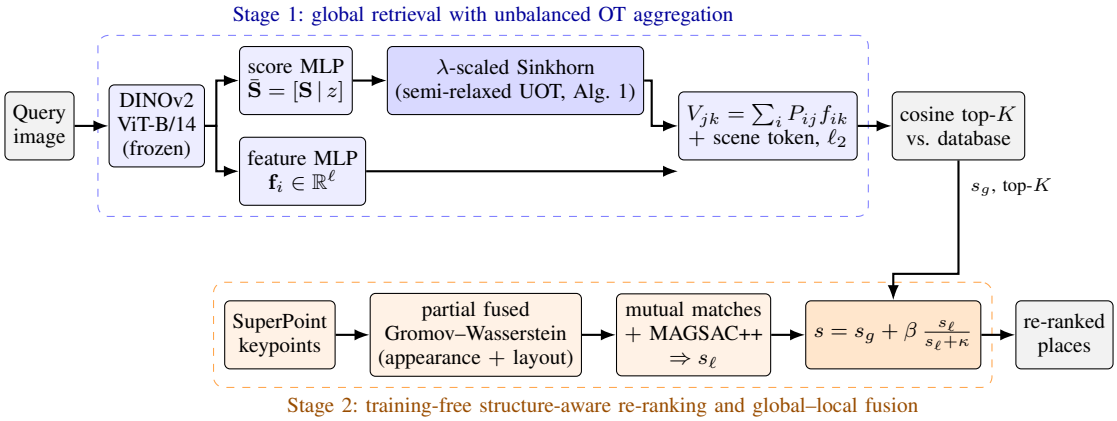



Fig. 1. Galatea. Stage 1 assigns ViT tokens to $m$ clusters and a dust-bin by semi-relaxed unbalanced OT with a learned relaxation $\lambda$ ($\lambda$=0: NetVLAD assignment, $\lambda$=1: SALAD), then aggregates them into a global descriptor. Stage 2 re-scores the top-$K$ candidates with a partial fused Gromov–Wasserstein matching of keypoints and fuses local and global scores.

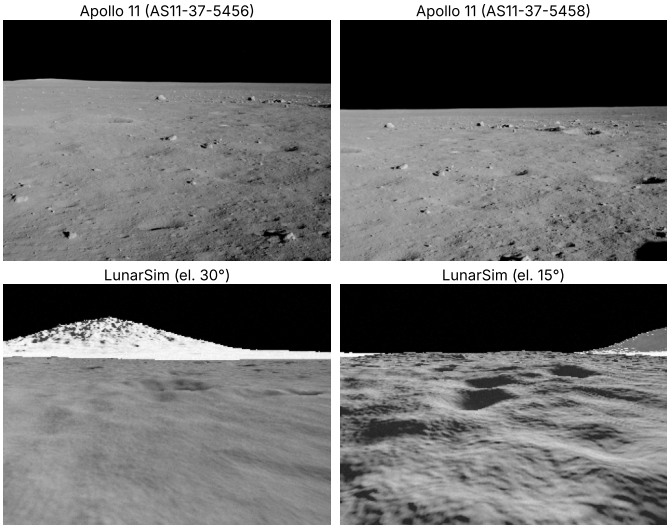



Fig. 2. Real Apollo 11 surface photographs [35] (top) and LunarSim renders (bottom). The renderer reproduces the black sky, the horizon structure and the shading statistics of low-Sun regolith. It under-resolves near-field rocks smaller than the 0.3 m DEM spacing, a limitation discussed in Sec. VII-A.

queries. For stage 2, the structural weight $\alpha \in \{0, 0.3, 0.6\}$ (chosen: 0.3) and the fusion parameters $\beta, \kappa$ are selected on the validation scene only. $\varepsilon_\ell = 0.05$ and the dust-bin cost threshold 0.25 (cosine 0.75) are fixed a priori, and MAGSAC++ uses a 1.5 px threshold. Fine-tuned heads are trained for 1000 steps with the same sampler and loss; the backbone learning rate ($10^{-4}$) was selected on the validation scene from $\{10^{-5}, \ldots, 10^{-3}\}$ with the Galatea head (supplement) and used for every head. The frozen-backbone results in Tables II and IV were computed on a 2-core CPU; fine-tuning and Table III used NVIDIA Tesla P40 GPUs, on which a re-run of all frozen-backbone heads reproduces the CPU results within seed noise (supplement). Code, the renderer, the rendered benchmark and the trained heads are available at https://github.com/RaunakDune/galatea-lunar-vpr.

### B. Terrestrial models under lunar illumination change

Table II (top) shows that all three terrestrial models perform reasonably when only the viewpoint changes: at C0, R@1 is 65–67%. They degrade sharply as soon as the Sun moves. DINOv2-SALAD, the strongest of them, falls from 66% at C0 to 21% under grazing illumination (C2) and to 25% when the shadows flip (C5). The CNN-based CosPlace and EigenPlaces fare worse, with C5 values of 10% and 16%. Even a foundation-model backbone pre-trained on 142M images therefore does not provide illumination-invariant place descriptors on regolith. Local re-ranking does not close the gap either. Fusing MNN+MAGSAC++ matches into the top-10 zero-shot SALAD candidates (selected on validation as in Sec. VI-E) raises R@1 to 51.1%. That is still far below any lunar-trained stage 1 *without* re-ranking, and only 31.9% on the hardest conditions, because the correct place is often not among the candidates at all.

### C. The assignment family with a frozen backbone

Training only the aggregation head on illumination-diverse LunarSim scenes lifts R@1 from 35.8% to 59.8% (Galatea head, three seeds; paired gain 24.0 points, 95% CI [22.4, 25.7]) and mean R@1 over the three hardest conditions (C2, C5, C6) from 23.4% to 47.6% (Table II). All cluster-based heads clearly beat every zero-shot model, whereas GeM pooling of the same frozen tokens does not (31.5%): global pooling cannot separate place-specific from illumination-dependent tokens [10].

Table III (left) confirms predictions (P1) and (P2). With a dust-bin, every value of $\lambda$ from 0 to 1 reaches the same R@1 within seed noise (spread across fixed $\lambda$: 1.5 points), and Galatea's learned head is statistically indistinguishable from SALAD (+0.4 points, CI $[-0.5, +1.3]$). A learned $\lambda$ barely leaves its initialisation ($\bar{\lambda} = 0.50$); even with a 25× larger learning rate on $\lambda$ and $z$ it only reaches 0.56, at unchanged accuracy (60.3%). This happens although the heads spend mass very differently: the balanced head sends 83.3% of the token mass to the dust-bin, an amount that tracks the sky fraction of the image ($\rho = 0.63$), the learned head only

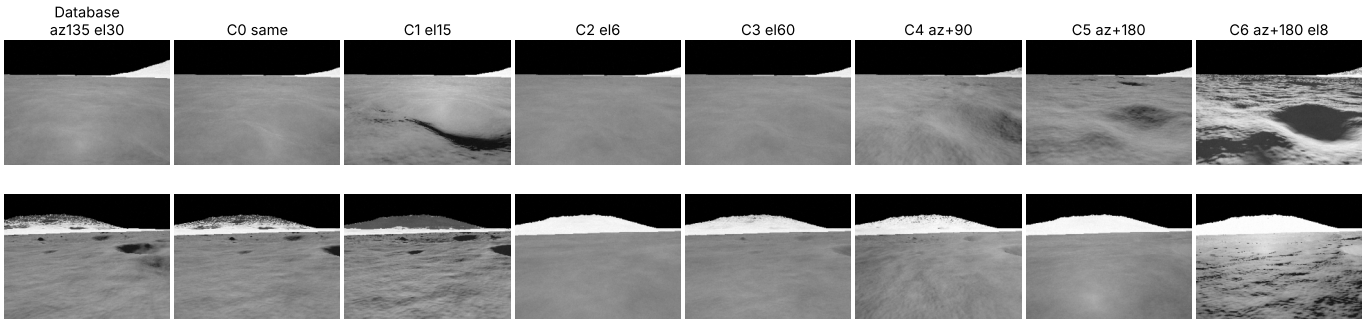



Fig. 3. LunarSim-VPR: two test places (rows). Left: database image (Sun azimuth $135°$, elevation $30°$). Right: queries at a jittered pose under the seven Sun conditions of Table I. Rendered with lunar-Lambert photometry, opposition surge and cast shadows from local and regional relief. In the top row at C2, the camera lies in the shadow of a distant massif and is lit only by the weak ambient term.

TABLE II
RECALL (%) ON THE LUNARSIM-VPR TEST SET (3 UNSEEN SCENES, 3150 QUERIES). LUNAR-TRAINED HEADS SHARE THE DINOV2 BACKBONE AND TRAINING PROTOCOL; *frozen*: ONLY THE HEAD IS TRAINED; *fine-tuned*: THE LAST FOUR BACKBONE BLOCKS ARE TRAINED JOINTLY WITH THE HEAD. MEAN±STD OVER 3 SEEDS. PER-CONDITION COLUMNS GIVE R@1 (CONDITIONS IN TABLE I).

| | Method | Dim. | R@1 | R@5 | R@10 | C0 | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| zero-shot | CosPlace-R50 [8] | 512 | 26.8 | 50.3 | 60.8 | 66.9 | 35.8 | 17.6 | 24.9 | 17.6 | 10.4 | 14.4 |
| | EigenPlaces-R50 [9] | 2048 | 28.1 | 53.0 | 64.1 | 64.9 | 34.2 | 17.1 | 25.3 | 22.2 | 15.6 | 17.1 |
| | DINOv2-SALAD [10] | 8448 | 35.8 | 65.4 | 77.7 | 66.4 | 46.7 | 21.3 | 31.6 | 35.8 | 24.9 | 24.0 |
| frozen | GeM | 1280 | 31.5±0.4 | 64.7±0.3 | 79.4±0.2 | 55.0 | 34.8 | 19.8 | 35.3 | 30.1 | 25.1 | 20.3 |
| | NetVLAD | 8448 | 58.2±0.7 | 86.6±0.4 | 94.6±0.2 | 79.0 | 67.5 | 46.1 | 65.2 | 56.0 | 48.3 | 45.6 |
| | OT, $\lambda$=0 | 8448 | 59.9±0.8 | 87.6±0.6 | 94.9±0.1 | 80.1 | 67.6 | 45.9 | 65.0 | 61.0 | 54.0 | 45.3 |
| | OT, $\lambda$=1 (SALAD) | 8448 | 59.4±0.7 | 87.3±0.7 | 94.8±0.2 | 77.6 | 68.4 | 46.4 | 65.4 | 58.6 | 52.7 | 46.7 |
| | OT, learned $\lambda$ | 8448 | 59.8±0.8 | 87.3±0.5 | 94.6±0.2 | 79.3 | 69.0 | 44.4 | 67.3 | 60.4 | 54.7 | 43.7 |
| fine-tuned | GeM | 1280 | 50.7±3.2 | 81.4±2.0 | 90.2±1.5 | 65.2 | 55.1 | 37.2 | 57.3 | 51.6 | 48.1 | 40.6 |
| | NetVLAD | 8448 | **76.4±0.7** | **94.3±0.1** | **97.6±0.2** | **91.6** | **85.3** | **63.0** | **86.1** | **78.9** | **71.8** | **58.1** |
| | OT, $\lambda$=0 | 8448 | 73.1±1.1 | 93.3±0.4 | 97.2±0.1 | 87.4 | 81.2 | 57.4 | 83.0 | 74.7 | 70.6 | 57.8 |
| | OT, $\lambda$=1 (SALAD) | 8448 | 73.5±0.5 | 94.0±0.2 | 97.4±0.2 | 88.5 | 80.1 | 57.7 | 83.0 | 76.4 | 71.3 | 57.6 |
| | OT, learned $\lambda$ | 8448 | 71.3±1.2 | 93.0±0.4 | 97.1±0.3 | 87.9 | 80.4 | 54.4 | 79.5 | 71.2 | 69.8 | 56.3 |

TABLE III
THE ASSIGNMENT FAMILY WITH A FROZEN AND A FINE-TUNED BACKBONE (TEST R@1, %, MEAN±STD OVER 3 SEEDS; HARD: MEAN R@1 OVER C2, C5, C6). ALL RUNS IN THIS TABLE WERE MADE ON THE SAME GPUS. THE OT ROWS SHARE THE AGGREGATION ARCHITECTURE AND DIFFER ONLY IN THE ASSIGNMENT; NETVLAD AGGREGATES RESIDUALS TO $k$-MEANS-INITIALISED CENTROIDS AND HAS NO DUST-BIN.

| | Frozen | | Fine-tuned | |
|---|---|---|---|---|
| Assignment | R@1 | hard | R@1 | hard |
| $\lambda = 0$, dust-bin | 60.1±0.9 | 48.1 | 73.1±1.1 | 61.9 |
| $\lambda = 0$, no dust-bin | 60.3±1.0 | 48.9 | 71.9±1.4 | 61.2 |
| $\lambda = 0.25$ | 60.0±1.1 | 48.1 | 73.0±1.4 | 62.4 |
| $\lambda = 0.5$ | 59.4±0.9 | 47.3 | 72.0±0.8 | 60.7 |
| learned $\lambda$ (Galatea, init. 0.5) | 59.5±1.3 | 47.4 | 71.3±1.2 | 60.1 |
| $\lambda = 0.75$ | 58.6±1.0 | 47.2 | 73.3±0.1 | 61.6 |
| $\lambda = 1$ (SALAD), dust-bin | 59.6±0.4 | 48.7 | 73.5±0.5 | 62.2 |
| $\lambda = 1$, no dust-bin | 55.8±1.5 | 44.1 | 74.4±0.9 | 62.9 |
| NetVLAD (residual, $k$-means) | 57.8±0.7 | 46.3 | 76.4±0.7 | 64.3 |

$0.4\%$ (coefficient of variation of cluster mass 6.4 vs. 0.18). As Prop. 4 states, intra-normalisation removes exactly this difference.

The dust-bin shows where the theory's assumptions bite. At $\lambda = 0$ it changes only the token weights $w_i$ and is irrelevant ($+0.1$ points without it, CI $[-0.5, +0.7]$). At $\lambda = 1$ it is essential ($-3.8$, CI $[-4.8, -2.7]$): without a place to put surplus mass, the balanced constraint forces every token, sky included, into the clusters in fixed proportions, which no re-weighting can undo. Residual NetVLAD aggregation is slightly below the OT heads on frozen tokens ($57.8\%$).

### *D. The assignment family with a fine-tuned backbone*

Fine-tuning the last four backbone blocks together with the head (Table II, bottom) improves every head substantially: GeM by 19.2, NetVLAD by 18.2, SALAD by 14.1 and the Galatea head by 11.5 points R@1 (CI $[10.2, 12.8]$; hard conditions: $47.6\% \rightarrow 60.1\%$). Within the OT family the spread stays small (Table III, right): the fixed values of $\lambda$ lie within 1.5 points of each other, and the dust-bin changes R@1 by about one point at most ($\lambda = 0$: $-1.2$ $[-2.1, -0.2]$; $\lambda = 1$: $+0.8$ $[-0.1, +1.8]$), presumably because the backbone learns to make uninformative tokens harmless. The learned head is the weakest member ($71.3\%$; $-2.2$, CI $[-3.2, -1.2]$ vs. SALAD), but the control with $\lambda$ fixed at its initial value 0.5 shows that learning is not the cause: $\lambda$ again stays near its initialisation ($\bar{\lambda} = 0.52$), and the learned head matches the fixed one ($72.0\%$; $-0.6$, CI $[-1.6, +0.3]$), as it does on a frozen backbone ($+0.2$, CI $[-0.3, +0.7]$). Learning $\lambda$ therefore neither helps nor hurts; it simply does not move away from a

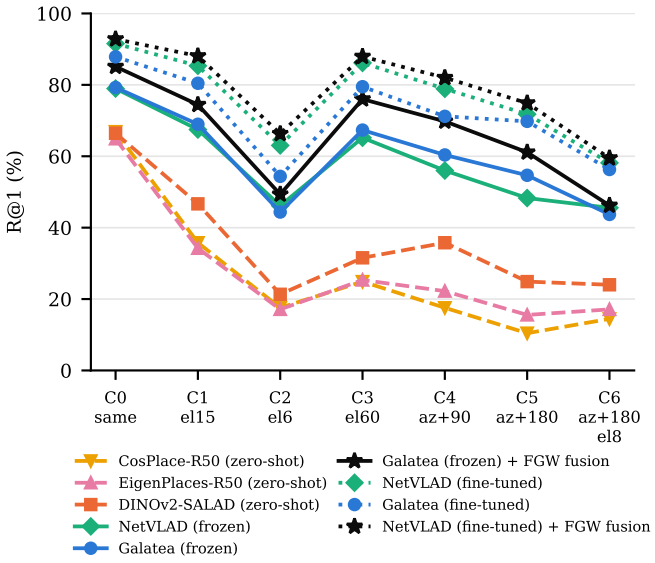



Fig. 4. R@1 per illumination condition on the test set. Zero-shot terrestrial models (dashed) collapse once the Sun moves. Lunar training of the aggregation head, global–local fusion and joint backbone fine-tuning (dotted) each recover part of the loss; grazing light (C2) and reversed grazing light (C6) remain the hardest conditions.

TABLE IV
STAGE-2 RE-RANKING OF THE TOP-10 GALATEA CANDIDATES (TEST, %). *Local*: SORT BY THE LOCAL SCORE ONLY. *Fused*: EQ. (12) WITH $(\beta, \kappa)$ CHOSEN ON THE VALIDATION SCENE.

| | Local | | Fused | |
|---|---|---|---|---|
| Local scorer | R@1 | R@5 | R@1 | R@5 |
| none (global ranking) | – | | 60.8 | 88.0 |
| MNN | 59.2 | 88.2 | 66.0 | 89.5 |
| MNN + MAGSAC++ | 62.4 | 89.2 | **67.2** | 90.0 |
| OT, appearance only | 57.0 | 89.5 | 65.8 | 90.4 |
| OT, appearance only + MAGSAC++ | 61.0 | 88.3 | 63.0 | 88.6 |
| LightGlue [22] (learned) | 57.5 | 87.4 | 65.3 | 89.6 |
| LightGlue [22] (learned) + MAGSAC++ | 60.0 | 89.0 | 65.7 | 89.8 |
| Partial FGW (ours) | 59.8 | 90.1 | 66.0 | 90.1 |
| Partial FGW (ours) + MAGSAC++ | 63.0 | 88.6 | 63.8 | 88.7 |

value that, with a fine-tuned backbone, happens to sit at the low end of a narrow range. Whether that dip at $\lambda = 0.5$ is real or seed noise cannot be decided with three seeds.

The largest difference is not inside the OT family at all. NetVLAD, which aggregates residuals to $k$-means-initialised centroids, becomes the best head ($76.4 \pm 0.7\%$). Its closest OT relative, $\lambda = 0$ without a dust-bin, uses the same softmax assignment and no dust-bin, yet is $+4.5$ points behind (CI $[+3.5, +5.5]$). With trainable tokens the *aggregation*, not the assignment, is therefore the design axis that matters; on frozen tokens the same comparison favours the OT head ($-2.4$ points for NetVLAD).

### E. Global–local fusion

Table IV re-ranks the top-10 candidates of the frozen Galatea head (seed 0, R@1 60.8%) with four local scorers. Two findings stand out. First, ranking by the local score alone barely helps and can hurt: no scorer improves R@1 by more than about two points, and several lose ground. Second, fusing local and global evidence with Eq. (12) helps every scorer. With partial FGW, R@1 rises to 66.0%, a gain of 5.2 points (95% CI $[4.1, 6.3]$), and on the hardest conditions from 47.8% to 52.2%, the best of all scorers on that subset. Differences between scorers in the fused setting are small and within the confidence intervals: FGW vs. LightGlue $+0.7$ $[-0.6, +2.0]$, and FGW vs. appearance-only OT $+0.2$ $[-0.8, +1.1]$. The training-free structural matcher thus performs as well as a learned terrestrial matcher. MAGSAC++ helps MNN and LightGlue but not the OT matchers, whose dust-bins already reject outliers.

TABLE V
WHY FUSION WORKS: SHARE OF TEST QUERIES (%) WHOSE TOP-1 RESULT RE-RANKING *rescues* (GLOBAL WRONG → CORRECT) OR *breaks* (GLOBAL CORRECT → WRONG), FOR LOCAL-ONLY RANKING AND FOR THE FUSED SCORE, ON GALATEA SEED 0.

| | | Local only | | Fused | |
|---|---|---|---|---|---|
| Stage 1 | Scorer | resc. | broken | resc. | broken |
| frozen | MNN+MAGSAC++ | 15.9 | 14.3 | 10.7 | 4.3 |
| | LightGlue | 14.2 | 17.6 | 9.3 | 4.8 |
| | partial FGW | 15.0 | 16.0 | 7.6 | 2.4 |
| fine-tuned | MNN+MAGSAC++ | 10.3 | 17.6 | 5.7 | 3.0 |
| | LightGlue | 10.0 | 21.0 | 4.4 | 1.8 |
| | partial FGW | 10.6 | 18.8 | 4.6 | 2.0 |

*Why local-only re-ranking fails and fusion works:* Table V splits the effect of re-ranking into queries it *rescues* (wrong global top-1, correct after re-ranking) and queries it *breaks* (the reverse). Local-only ranking rescues many queries, 15.0% with FGW, but breaks as many (16.0%): SuperPoint responses on regolith are dominated by shadow edges, so a wrong candidate photographed under similar light can collect more matches than the right one. The overrides that break a correct result are those against a confident global ranking: the median global similarity margin they overturn is 0.106, twice that of the overrides that rescue a query (0.050). This is the case that Prop. 6 rules out. Fusion keeps about half of the rescues (7.6%) while breaking only 2.4%, and the same pattern holds for every scorer and for the fine-tuned stage 1, where local-only ranking breaks up to 21.0% of the queries and fusion at most 3.0%.

*Robustness across stage-1 models:* Repeating the protocol on all three seeds of each stage-1 model, and re-tuning $\beta, \kappa$ on validation each time, fusion with FGW adds $6.3 \pm 0.8$ points to the frozen Galatea head (MNN+MAGSAC++: $6.9 \pm 0.4$; plain MNN: $5.2 \pm 0.6$), $3.1 \pm 0.4$ to the fine-tuned Galatea head (MNN+MAGSAC++: $3.3 \pm 0.6$) and $2.4 \pm 0.2$ to the fine-tuned NetVLAD head (MNN+MAGSAC++: $1.9 \pm 0.2$). The gain shrinks as stage 1 improves, because fewer queries remain to be rescued, but it never vanishes. The complete system, fine-tuned NetVLAD with FGW fusion, reaches $78.8 \pm 0.5\%$ R@1 and 95.3% R@5 (66.9% on the hardest conditions), against 35.8% for the best terrestrial model. On the Moon, *how local and global evidence are combined matters more than which local matcher is used.*

TABLE VI
ABLATIONS OF THE GALATEA HEAD (TEST, %, SEED 0; HARD = MEAN R@1 OVER C2, C5, C6). SINGLE-ILLUMINATION: THE TRAINING SCENES RE-RENDERED WITH THE SAME POSES UNDER THE DATABASE SUN ONLY.

| Variant | R@1 | R@5 | hard |
|---|---|---|---|
| Frozen backbone, reference | 60.8 | 88.0 | 47.8 |
| $L = 1$ Sinkhorn iteration | 59.9 | 87.8 | 46.6 |
| $L = 10$ iterations | 60.7 | 88.0 | 47.7 |
| $m = 16$ clusters | 57.7 | 86.2 | 45.9 |
| $m = 32$ clusters | 61.1 | 88.2 | 47.9 |
| single-illumination training | 47.3 | 72.2 | 27.6 |
| SALAD, single-illumination | 46.9 | 72.6 | 28.5 |
| Fine-tuned backbone, reference | 72.8 | 93.6 | 60.9 |
| single-illumination training | 51.2 | 73.7 | 24.7 |
| SALAD, single-illumination | 54.0 | 76.0 | 29.6 |

### F. Training illumination and other ablations

Table VI varies one factor at a time on seed 0. Single-seed differences below about two points should be read against the seed standard deviation of roughly one point (Table II).

*Illumination-diverse training:* Re-rendering the five training scenes with the *same* poses but a single Sun direction (that of the database) has by far the largest effect of any factor we varied. The frozen Galatea head then reaches only $47.3\%$ R@1, $13.6$ points below illumination-diverse training, and SALAD drops the same way ($46.9\%$). The single-illumination model is slightly *better* when the query light matches its training light (C0: $85\%$ vs. $80\%$), but on the three hardest conditions it falls to $27.6\%$, against $47.8\%$. Fine-tuning makes this worse, not better: with a single training Sun the fine-tuned Galatea reaches $51.2\%$ ($-21.5$ points; C0 $93\%$ vs. $89\%$, hard $24.7\%$ vs. $60.9\%$) and SALAD $54.0\%$. A more flexible model fits the training light more closely, so backbone adaptation is only useful together with illumination diversity. For lunar VPR, an illumination sweep in the training data, as provided by renderers such as LunarSim or DEM-based pipelines, matters far more than the choice of aggregation layer.

*Sinkhorn iterations and clusters:* Changing the number of Sinkhorn iterations ($L = 1$: $59.9\%$, $L = 10$: $60.7\%$) moves R@1 by less than seed noise, in line with prediction (P3) and the fast contraction of Prop. 2. Halving the clusters to $m = 32$ (4352-D) costs nothing ($61.1\%$), which is attractive for bandwidth-limited rovers. With $m = 16$ (2304-D), R@1 drops to $57.7\%$.

### G. Run time

On the same 2-core CPU, the frozen ViT-B/14 takes 228 ms per image and SuperPoint 140 ms. The heads add only 1.8–3.0 ms (Galatea: 3.0 ms), so the $\lambda$-scaled Sinkhorn loop is negligible. Per candidate pair (200 keypoints), appearance-only OT takes 8.3 ms, LightGlue 19 ms and partial FGW 43 ms ($\approx 427$ ms per query for $K = 10$), so MNN+MAGSAC++ is the cheaper choice at similar accuracy. On a Tesla P40 GPU the backbone takes 15 ms per image.

## VII. DISCUSSION: WHAT MATTERS FOR LUNAR VPR

Viewing both stages as OT problems lets us separate three kinds of design decision and rank them by their measured effect on LunarSim-VPR.

*Training data first:* Illumination diversity in training is the largest single factor: it is worth $13.6$ points of R@1 with a frozen backbone and $21.5$ with a fine-tuned one, more than any architectural choice. Because a more adaptable model fits a single training light more closely, illumination diversity and backbone fine-tuning have to go together.

*Then backbone adaptation, then aggregation:* Fine-tuning the last four blocks adds 11–19 points for every head. The aggregation matters only at the coarse level (cluster-based aggregation beats global pooling by more than 20 points) and, once the backbone is trained, at the level of residual vs. OT aggregation. Within the OT family the assignment is a weak design lever: all values of $\lambda$ are equivalent on a frozen backbone (P1, P2), and on a fine-tuned backbone the fixed members stay within $1.5$ points of each other while a learned $\lambda$ behaves like a fixed $\lambda$ at its initial value. The cheapest member, $\lambda = 0$, needs no Sinkhorn loop and matches SALAD's balanced assignment within seed noise. A learnable $\lambda$ adds a parameter that the loss barely constrains (P2), so it should be fixed; $\lambda = 0$ is the natural choice for speed, and no fixed value is clearly better than another.

*Fusion, not the matcher:* Local evidence on the Moon is abundant but unreliable, because shadow edges dominate keypoint detections. Its value lies in the queries where the global ranking is uncertain, which is exactly where the bounded-override property of the fusion rule (Prop. 6) lets it act. With fusion in place, a training-free OT matcher, plain mutual nearest neighbours and a learned terrestrial matcher perform alike, so the cheapest adequate scorer, MNN+MAGSAC++, is a sensible default (LightGlue costs $\approx 192$ ms and FGW $\approx 427$ ms per query on a CPU). The structural FGW matcher earns its cost mainly where no learned matcher is available or trusted.

*What the theory did and did not predict:* Prop. 4 correctly predicted that, with fixed tokens, the assignment family collapses to one operating point and that a learned $\lambda$ would not move. It does not predict the ordering once the tokens are trained; that ordering, and the advantage of residual aggregation, are empirical findings on simulated data that should be re-examined on real imagery.

### A. Limitations

All quantitative results come from simulation. LunarSim reproduces the statistical character of lunar relief and the dominant photometric effects, but it omits several things: resolved near-field rocks below the $0.3$ m DEM spacing (Fig. 2), multiple scattering beyond a constant ambient term, rover self-shadowing and dust, lens flare, exposure bracketing, and the full range of geological units. Absolute recall values are therefore not predictions of on-Moon performance. The *relative* comparisons use identical conditions and paired queries on unseen scenes, so we expect them to carry over better, but this must be confirmed on real imagery. Fine-tuning used a

single learning rate selected on one validation scene and four of twelve blocks; full fine-tuning or larger backbones were not explored, and the ranking of heads under fine-tuning may depend on these choices. The FGW structural term uses image-plane distances, which are only approximately preserved under large viewpoint changes. Lifting database keypoints to 3-D with DEM depth is a natural extension.

## VIII. Conclusion

Galatea treats lunar visual place recognition as optimal transport at both stages: a family of semi-relaxed unbalanced assignments for global aggregation, which contains the NetVLAD and SALAD assignments and a learned relaxation, and a partial fused Gromov–Wasserstein matcher whose evidence is fused with the global score by a rule with bounded override. The theory of the assignment family predicted, and LunarSim-VPR confirmed, that its members are equivalent on a frozen backbone and that a learned relaxation would not move; once the backbone is fine-tuned the family spreads only slightly, learning $\lambda$ brings no benefit over fixing it, and residual aggregation is best. Global–local fusion improved every stage-1 model and every local matcher, because it keeps about half of what local evidence rescues without overturning confident global decisions. Illumination-diverse training remained the largest single factor. Next steps are real imagery, with DESPINA-synthesised reference views of real sites [31], and 3-D structural terms from DEM depth.

## References


[1] M. Maimone, Y. Cheng, and L. Matthies, "Two years of visual odometry on the Mars exploration rovers," *Journal of Field Robotics*, vol. 24, no. 3, pp. 169–186, 2007.

[2] A. E. Johnson and J. F. Montgomery, "Overview of terrain relative navigation approaches for precise lunar landing," in *Proc. IEEE Aerospace Conference*, 2008.

[3] M. Ceresoli, S. Silvestrini, and M. Lavagna, "Optical image generation through digital terrain models for autonomous lunar navigation," *Aerospace*, vol. 12, no. 2, p. 92, 2025.

[4] S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford, "Visual place recognition: A survey," *IEEE Transactions on Robotics*, vol. 32, no. 1, pp. 1–19, 2016.

[5] S. Schubert, P. Neubert, S. Garg, M. Milford, and T. Fischer, "Visual place recognition: A tutorial," *arXiv preprint arXiv:2303.03281*, 2023.

[6] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.

[7] F. Radenović, G. Tolias, and O. Chum, "Fine-tuning CNN image retrieval with no human annotation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 7, pp. 1655–1668, 2019.

[8] G. Berton, C. Masone, and B. Caputo, "Rethinking visual geo-localization for large-scale applications," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022.

[9] G. Berton, G. Trivigno, B. Caputo, and C. Masone, "EigenPlaces: Training viewpoint robust models for visual place recognition," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023.

[10] S. Izquierdo and J. Civera, "Optimal transport aggregation for visual place recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 17 658–17 668.

[11] S. Hausler, S. Garg, M. Xu, M. Milford, and T. Fischer, "Patch-NetVLAD: Multi-scale fusion of locally-global descriptors for place recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 14 141–14 152.

[12] G. Neukum, B. A. Ivanov, and W. K. Hartmann, "Cratering records in the inner solar system in relation to the lunar reference system," *Space Science Reviews*, vol. 96, pp. 55–86, 2001.

[13] H. J. Melosh, *Impact Cratering: A Geologic Process*. Oxford University Press, 1989.

[14] A. Torii, J. Sivic, T. Pajdla, and M. Okutomi, "Visual place recognition with repetitive structures," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 883–890.

[15] A. S. McEwen, "Photometric functions for photoclinometry and other applications," *Icarus*, vol. 92, no. 2, pp. 298–311, 1991.

[16] B. Hapke, *Theory of Reflectance and Emittance Spectroscopy*, 2nd ed. Cambridge University Press, 2012.

[17] H. Jégou, M. Douze, C. Schmid, and P. Pérez, "Aggregating local descriptors into a compact image representation," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2010, pp. 3304–3311.

[18] A. Ali-bey, B. Chaib-draa, and P. Giguère, "MixVPR: Feature mixing for visual place recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2023, pp. 2998–3007.

[19] N. Keetha, A. Mishra, J. Karhade, K. M. Jatavallabhula, S. Scherer, M. Krishna, and S. Garg, "AnyLoc: Towards universal visual place recognition," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1286–1293, 2024.

[20] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.

[21] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.

[22] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.

[23] D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperPoint: Self-supervised interest point detection and description," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2018, pp. 224–236.

[24] F. Mémoli, "Gromov–Wasserstein distances and the metric approach to object matching," *Foundations of Computational Mathematics*, vol. 11, no. 4, pp. 417–487, 2011.

[25] G. Peyré, M. Cuturi, and J. Solomon, "Gromov-Wasserstein averaging of kernel and distance matrices," in *Proc. Int. Conf. Machine Learning (ICML)*, 2016, pp. 2664–2672.

[26] T. Vayer, L. Chapel, R. Flamary, R. Tavenard, and N. Courty, "Optimal transport for structured data with application on graphs," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6275–6284.

[27] L. Chapel, M. Z. Alaya, and G. Gasso, "Partial optimal transport with applications on positive-unlabeled learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[28] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2013, pp. 2292–2300.

[29] G. Peyré and M. Cuturi, "Computational optimal transport," *Foundations and Trends in Machine Learning*, vol. 11, no. 5–6, pp. 355–607, 2019.

[30] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced optimal transport problems," *Mathematics of Computation*, vol. 87, no. 314, pp. 2563–2609, 2018.

[31] A. Nelson-Archer, R. Sarbajna, and C. F. Eick, "DESPINA: Synthesis of high-fidelity planetary horizon reconstructions using DEM-guided diffusion," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XI-2-2026, pp. 881–890, 2026.

[32] Y. Zhang, Z. Kang, and Z. Cao, "An image retrieval method for lunar complex craters integrating visual and depth features," *Electronics*, vol. 13, no. 7, p. 1262, 2024.

[33] X. Wang, X. Han, W. Huang, D. Dong, and M. R. Scott, "Multi-similarity loss with general pair weighting for deep metric learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5022–5030.

[34] D. Barath, J. Noskova, M. Ivashechkin, and J. Matas, "MAGSAC++, a fast, reliable and accurate robust estimator," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 1304–1312.

[35] NASA, "Apollo 11 image library," https://www.nasa.gov/history/alsj/a11/images11.html, 2020.

[36] A. Ali-bey, B. Chaib-draa, and P. Giguère, "GSV-Cities: Toward appropriate supervised visual place recognition," *Neurocomputing*, vol. 513, pp. 194–203, 2022.